Ryan Peckham

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Education

University of Michigan — Ann Arbor, MI
B.S. Computer Engineering and Applied Physics; GPA: 3.35/4.0 — Aug. 2025 – Present
Sophomore – Anticipated Graduation May 2029
Coursework: Multivariable & Vector Calculus, Electricity & Magnetism, Quantum Photonics, Discrete Mathematics, Solar Energy Systems, Mandarin, Probabilistic Statistics

Skills

Programming Languages: Python, Rust, C, C++, C#, Java, SQL, Front-End Stack, MATLAB
Systems: Git & GitHub, PostgreSQL, WSL, Docker, Database Normalization, Threading
Operating Systems: Windows, Linux (Ubuntu, Debian, NixOS), MacOS
Hardware: Soldering, Shielded Metal Arc Welding, CNC Machines, Lathes, Mills

Experience

Michigan Autonomous Security Team (M-AST), Chief Software Engineer — Jan. 2026 – Present
University of Michigan
- Leading and instructing a team of engineers to install, program, and test gyroscope, magnetometer, barometer, and GPS sensors onto a prototype drone
- Maintaining the organization of a large and complex codebase and reviewing code and pull requests
- Collaborated with M-AST President and peers to design the drone architecture specifying the flow of information between systems, including the implementation of SITL and HITL runtimes
- Integrated several software architectures including ROS2, OpenCV, telemetry code, and path-finding

Michigan Hackers, Machine Learning Team, Member — Nov. 2025 – Present
University of Michigan
- Developed an understanding of fundamental machine learning concepts including Markov Chains, Baye's Theorem, and various machine learning metholodogies and applications
- Led and participated in peer lectures and collaborative discussions regarding the modern development and usage of AI in technology

Engineering 100 — Aug. 2025 – Dec. 2025
University of Michigan
- Completed the C++ programming, wiring, and soldering of a sun-tracking solar panel using photoresistors, stepper motors, voltage limiters, and a battery on a Raspberry Pi platform
- Developed skills in communication, collaboration, task prioritization, concluding meaningful results from data, and balancing planning with execution

FIRST Robotics Competition (FRC), Team 141 "WOBOT" — Aug. 2023 – Oct. 2024
West Ottawa High School, Programming Sub-team
- Developed input and movement code by implementing and tuning PIDF loops for precise control
- District Champions; awards won by team: Engineering Inspiration, Sustainability, Entrepreneurship

Cultural, Technological, and Environmental Exchange (CTEE) Club — Sep. 2024 – Apr. 2025
West Ottawa High School
- Repaired and tested 2–3 community-donated desktop computers and monitors each week
- Learned about the architecture and design of desktop computers & motherboards
- Replaced faulty components safely and prepared ready-to-use operating systems (Windows)
- Stored repaired units for future shipment to developing nations and communities

Personal Projects

- Designed and developed my personal website peckhamr.com using HTML, JS, CSS, and the Svelte UI framework, as an opportunity to teach myself the foundations of domain registrars, website hosting, and maintainable codebase design
- Applied the skills I learned to create the public website for M-AST, flymast.com; a more professional and sleek design built for prospective students and sponsors

Work Experience

Ace Hardware, Delivery and Repair Technician — May 2026 - Aug. 2026
Chick-fil-A, Crew Member — Feb. 2024 - Jan. 2026
Target, Electronics Consultant & Team Leader — Nov. 2020 - Aug. 2022